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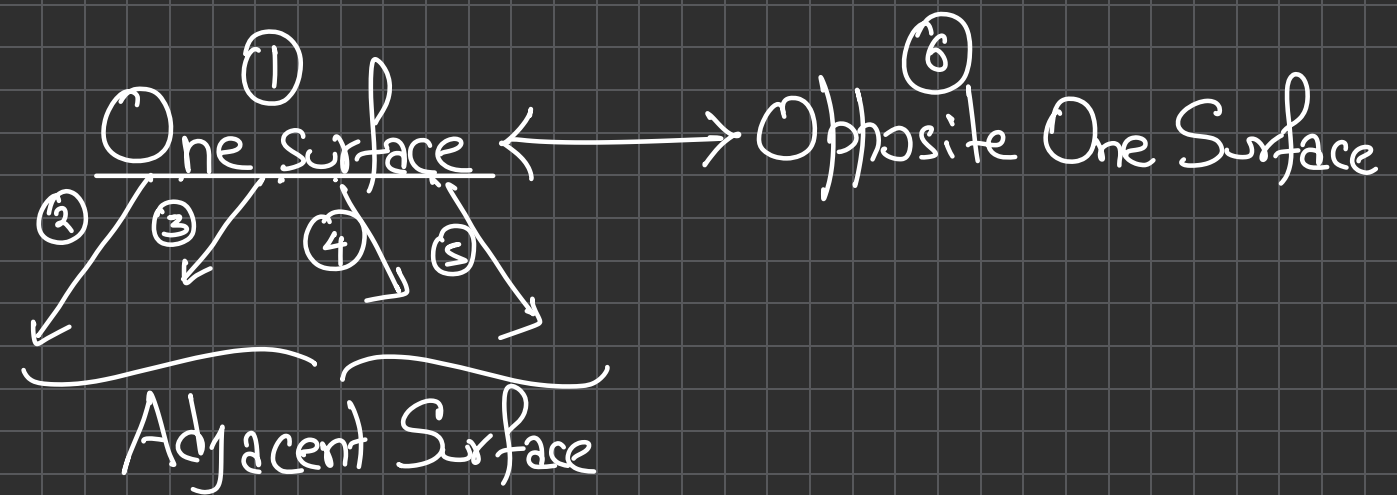
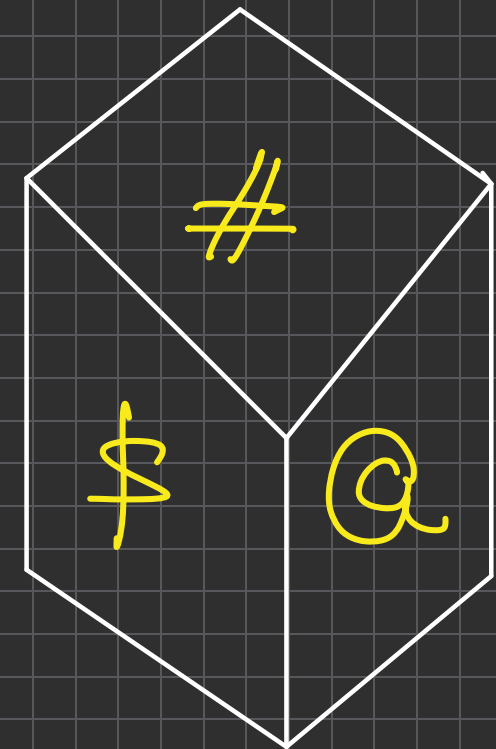
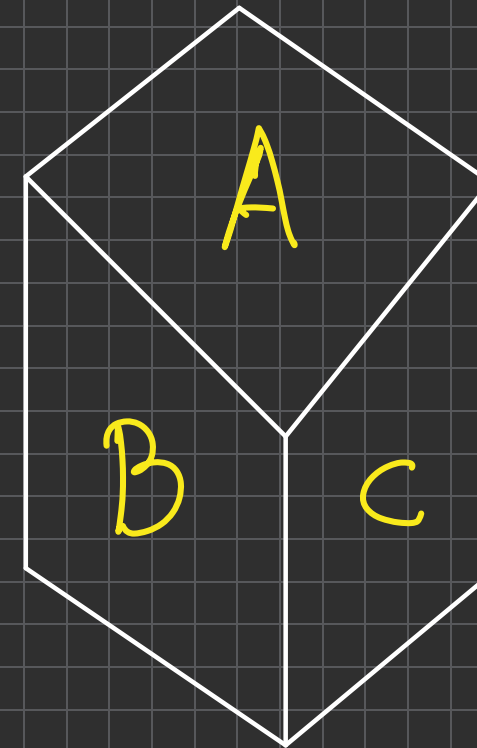
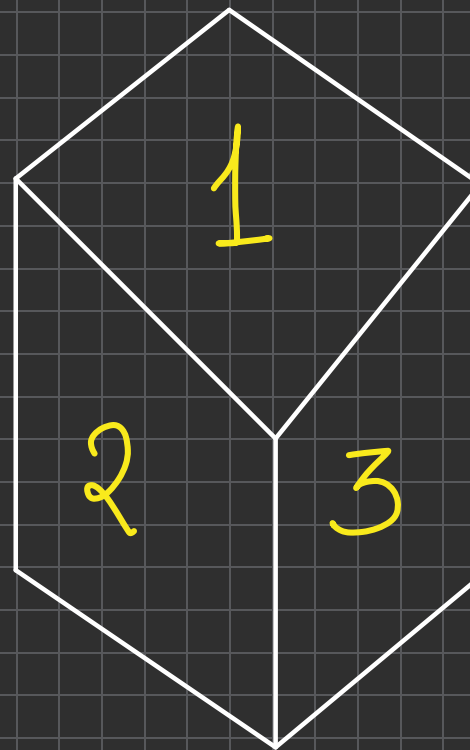
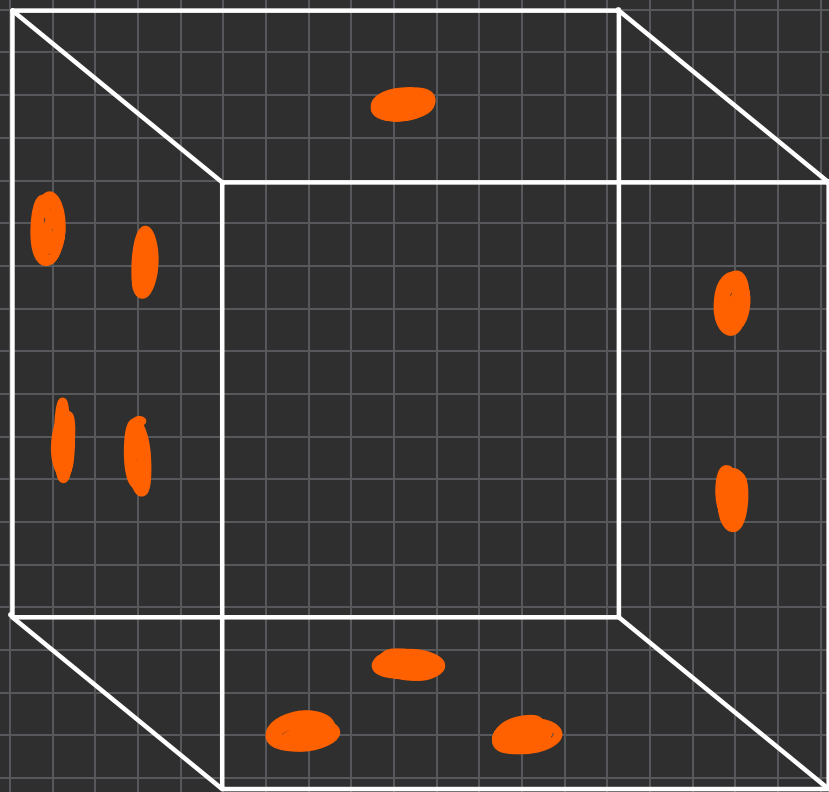
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Dice



Types of Dice

1) Standard Dice $\left\{ \begin{array}{l} 1-6 \\ 2-5 \\ 3-4 \end{array} \right\}$

If in the dice sum of opposite surface is always 7.

2) Non-Standard Dice

If in the dice, sum of any 2 adjacent side is 7

Q) If 3 standard dice thrown on the ground & sum of the upper surface sum is showing 13.

What is the sum of the number which comes on the lower surface of the dice

A) 9 B) 12 C) 8 D) 6

As per the formula, two opposite side sum = 7

$$3 \text{ dice} = 7 + 7 + 7 = 21$$

Upper Surface Sum = 13

$$\therefore \text{Lower surface} = 21 - 13 = \underline{\underline{8}}$$

Option C

3 Fundamental Rules of Non-Standard Dice

1) Adjacent Rule - In the dice, adjacent number never come on the opposite surface

2) Common Rule - If two numbers are common in two diagrams of dice, then remaining two numbers is always opposite to each other

Q) 4 standard dice thrown on the ground, if sum of the numbers on the upper surface is 15. What is sum of the number on the lower surface.

A) 6 B) 13 C) 12 D) 3

As per the formula, sum of two opposite side = 7

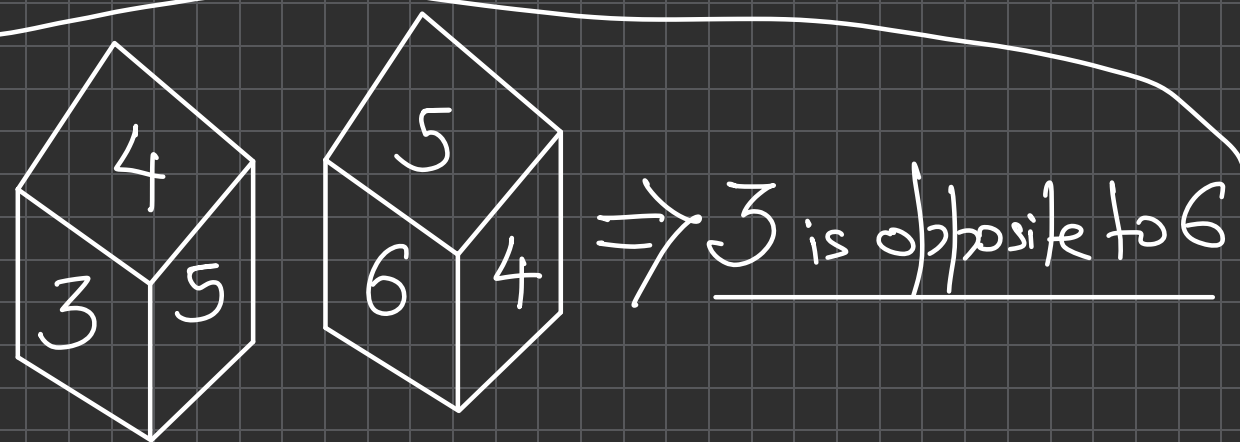
$$4 \text{ Dice} = 7 + 7 + 7 + 7 = 28$$

Given sum of 4 upper surface of dice = 15

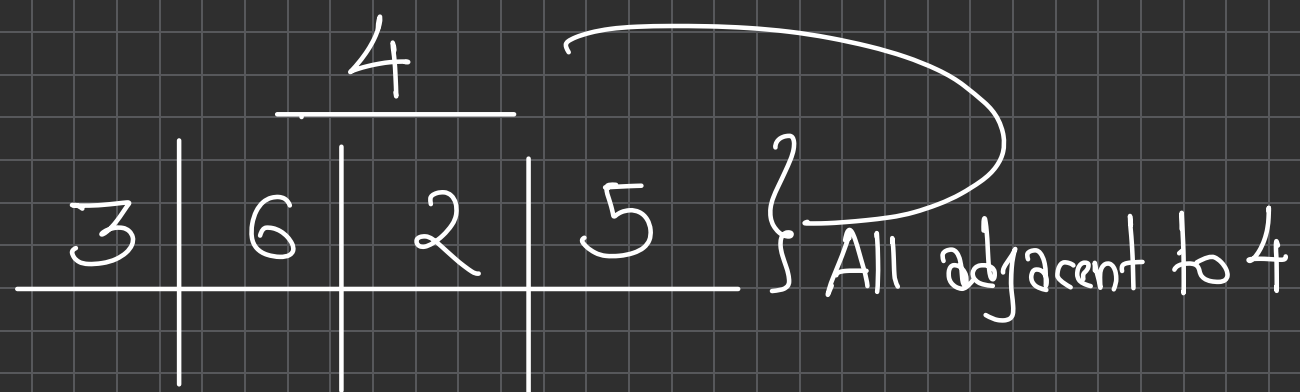
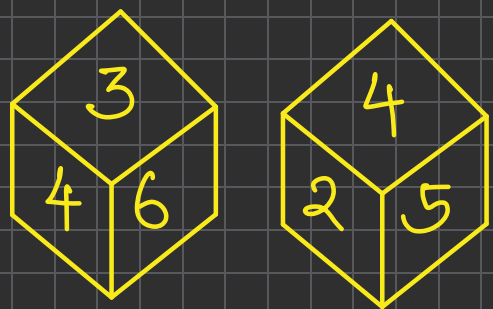
$$\therefore \text{Lower surface} = 28 - 15 = \underline{\underline{13}}$$

Option B

3) Rotational Rule - If only one number is the two diagram of dice, rotate clockwise with the common number of two diagram of dice

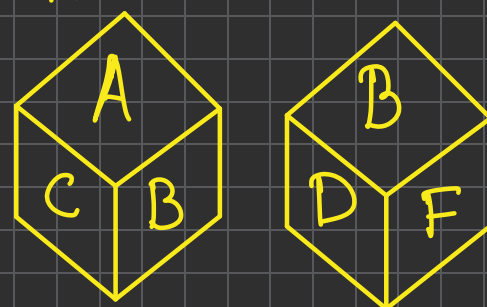


Q) Give two images of the same non-standard dice
Find the number opposite to 4



∴ Only 1 is remaining

Q) In the given diagram which alphabet is opposite to B.

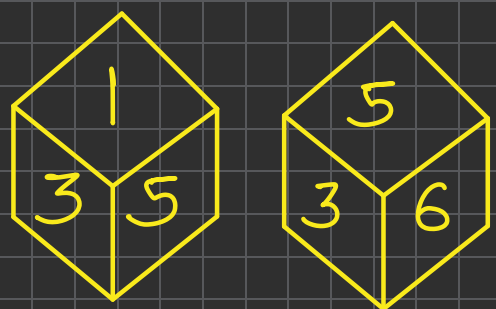


A) E
B) C
C) A
D) F

Adjacent to B $\Rightarrow$ A, C, D, F
Remains E $\Rightarrow$ Option A

Note: Any dice not having digits on its surface is always a Non-Standard Dice

Q) In the given diagram which number is opposite to 6.

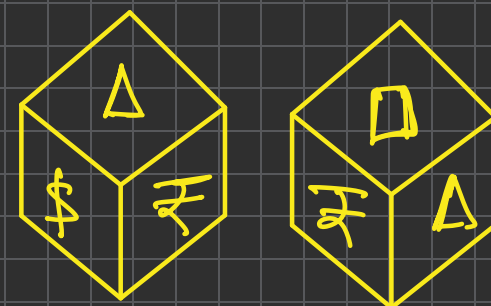


A) 1
B) 2
C) 4
D) 3

As per the 2nd Fundamental Rule
3, 5 were common therefore

$1 \Leftrightarrow 6$, Option A

Q) In the given diagram which symbol is opposite to \$

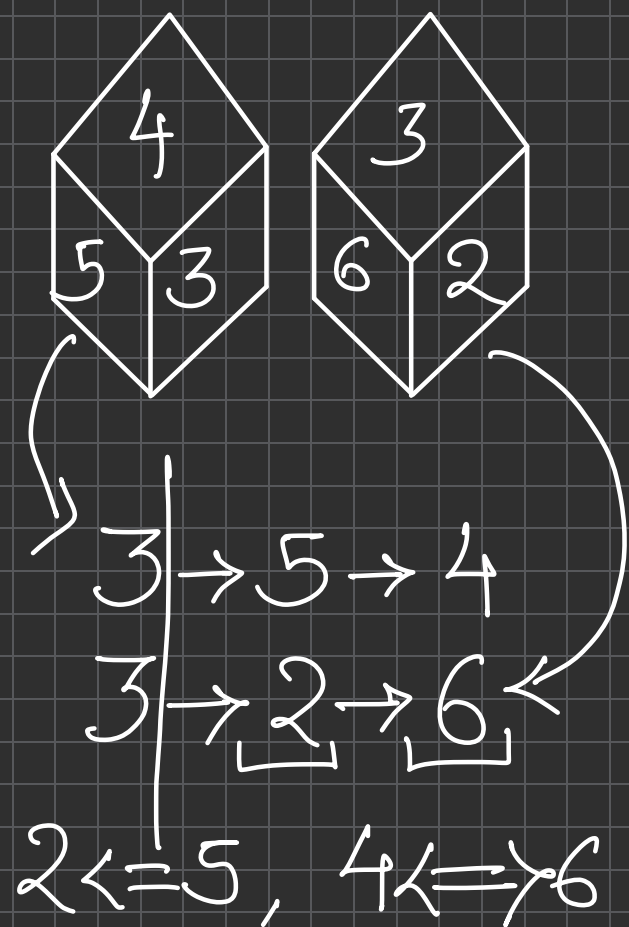


A) €
B) □
C) ↑
D) Δ

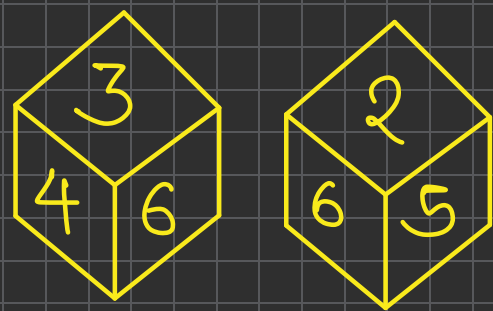
As per the 2nd Fundamental Rule

₹, Δ are adjacent, therefore $\$ \Leftrightarrow \square$, B

3. Rotational Rule



Q In the given diagram which number is opposite to 5.



- A) 1
- B) 2
- C) 3
- D) 4

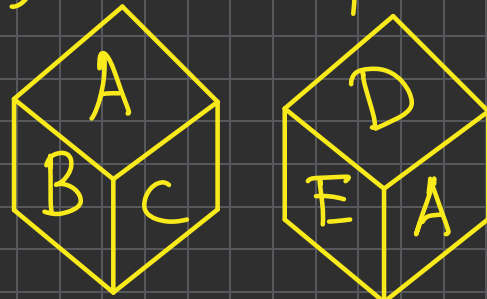
As per the 3rd fundamental Rule
6 is Common.

6 $\rightarrow$ 4 $\rightarrow$ 3

6 $\rightarrow$ 2 $\rightarrow$ 5

2 $\leftrightarrow$ 4, 3 $\leftrightarrow$ 5 $\rightarrow$ Option C

Q Which alphabet is opposite to C



- A) D
- B) E
- C) F
- D) C

As per the 3rd fundamental Rule
A is Common

A $\rightarrow$ C $\rightarrow$ B $\therefore$ C $\leftrightarrow$ E

A $\rightarrow$ E $\rightarrow$ D
 Option B

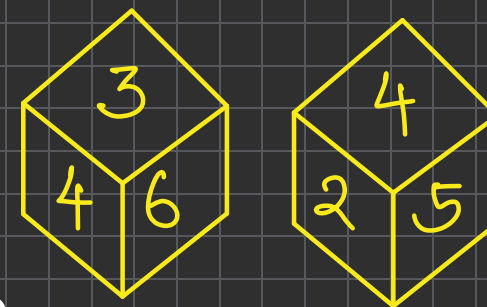
Note:

When a common number is asked, then apply Rule 1. Adjacent Rule

If the number is not a common number, then apply Rotational Rule

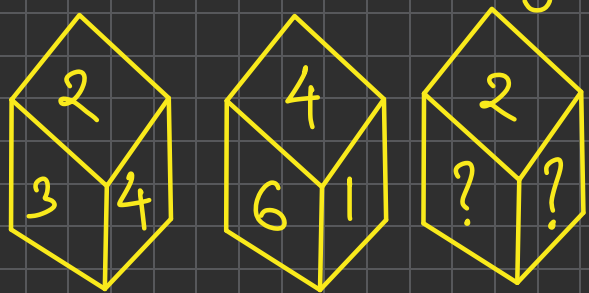
In Rule 2. Common Rule, 2 numbers are common so cannot be used.

For example



Q To find a number opposite
 $\rightarrow$ 4 $\rightarrow$ Apply Rule 1 (4 $\leftrightarrow$ 1)
 $\rightarrow$ 5 $\rightarrow$ Apply Rule 3 (3 $\leftrightarrow$ 5)

Q) The number adjacent of 2.



- A) 3 & 6
- B) 5 & 6
- C) 3 & 5
- D) 6 & 4

Apply rotational rule, keeping 4 as axis

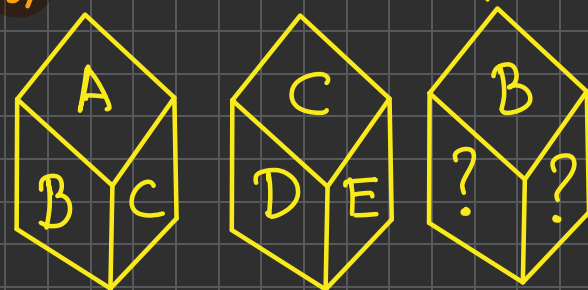
4 → 3 → 2

4 → 1 → 6

As 6 is opposite to 2.

∴ Adjacent to 2 → 1, 3, 4, 5
Option C

Q) Find the '?' Alphabets



- A) E & A
- B) C & E
- C) F & C
- D) D & F

Apply Rotational Rule on C.

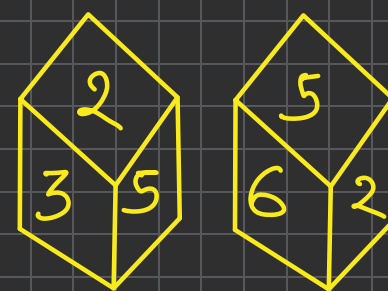
C → B → A } B ↔ E

C → E → D }

∴ Adjacent to B: A, C, D, F

A
C [B] F } Option D
D

Q) Find the number opposite to 2



Applying adjacent rule
We can say
3, 5, 6 are not
opposite.

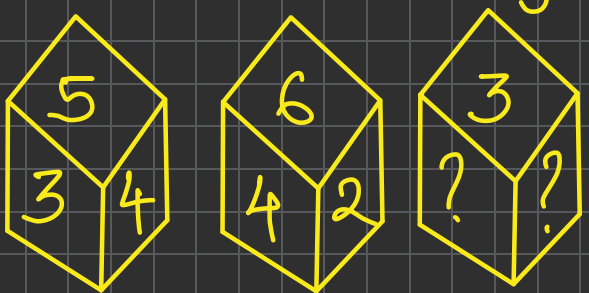
But not sure for 1 & 4.

∴ Ans - 1/4

Note: In case of Non-standard dice the
correct opposite cannot be determined but
in Standard Dice we can find.

(1-6), (2-5), (3-4)

Q) The number adjacent of 3



- A) 6 & 1
- B) 4 & 1
- C) 5 & 2
- D) 2 & 1

Apply Rotational rule on 4

4 → 3 → 5

4 → 6 → 2

4 [3] 1 } Option D
5

Open Dice

Generally there are 6 patterns of open dice

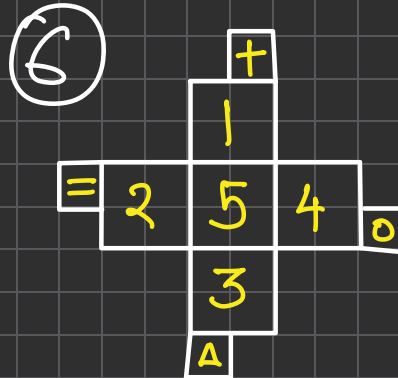
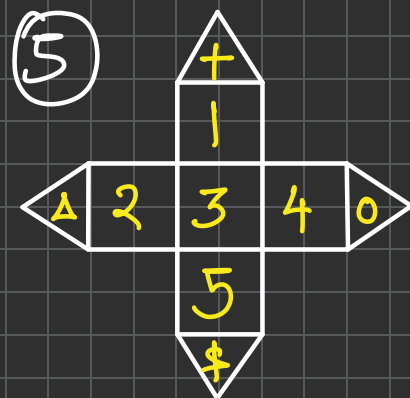
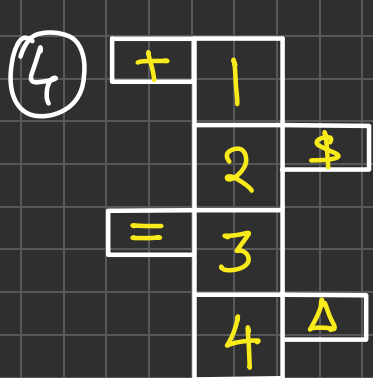
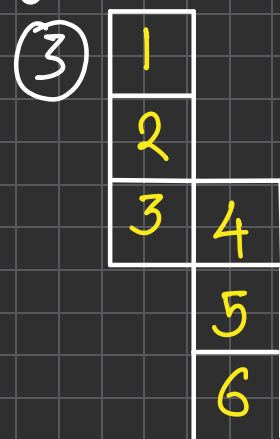
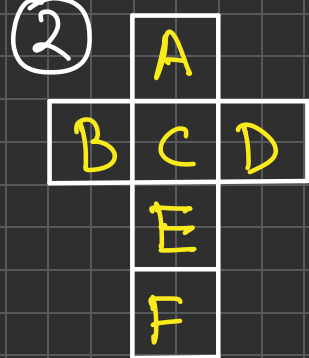
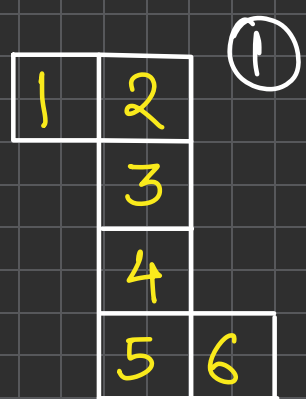


Diagram 4.

1-3, 2-4

+, \$, =, Δ ← Half surface

∴ 1-3 ⇒

+	=
---	---

2-4 ⇒

\$	Δ
----	---

Diagram 5

2-4

1-5

∴ 3 →

+	Δ
\$	0

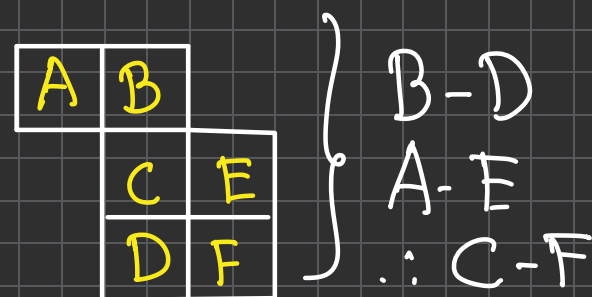
Diagram 6

1-3, 2-4

∴ 5 ⇒

=	+
Δ	0

Another example of open dice



Rule for open dice

→ In the open dice, opposite surface always exist in straight line in the gap of 1 box

In Diagram ①

2-4, 3-5

opposite

∴ 1-6

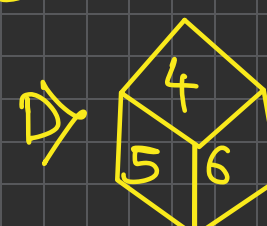
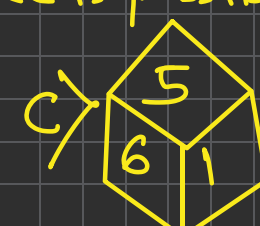
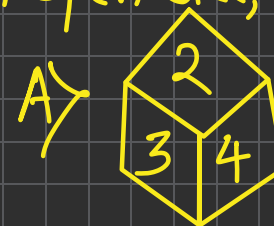
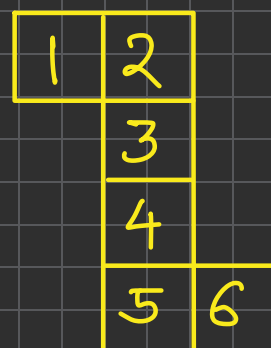
In Diagram ②

A-E, B-D, C-F

Diagram ③

1-3, 4-6, ∴ 2-5

Q From the given open dice, which close dice is possible



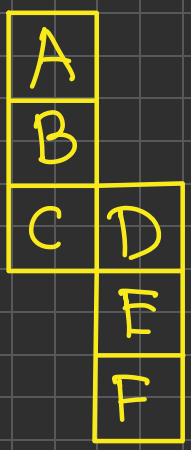
→ 2-4 }
 3-5 } →

2
3 5
4

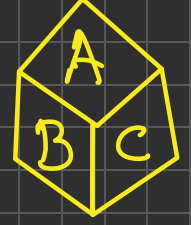
 6
 ∴ 1-6

Option D

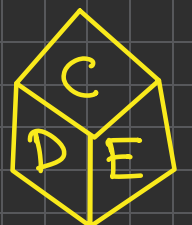
Q



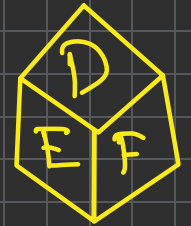
A



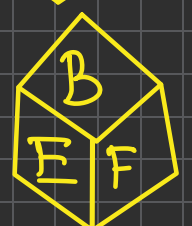
B

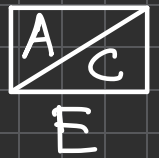


C



D



A-C
D-F
∴ B-E } → F  D Option B